

 Italy — Monthly Intelligence Report v4.2

Edition 006 — The Puglia Corridor

Where the energy transition is outrunning the grid. edition · August 2026 · SSI v4.2  
(refreshed June 2026)

CONTENTS

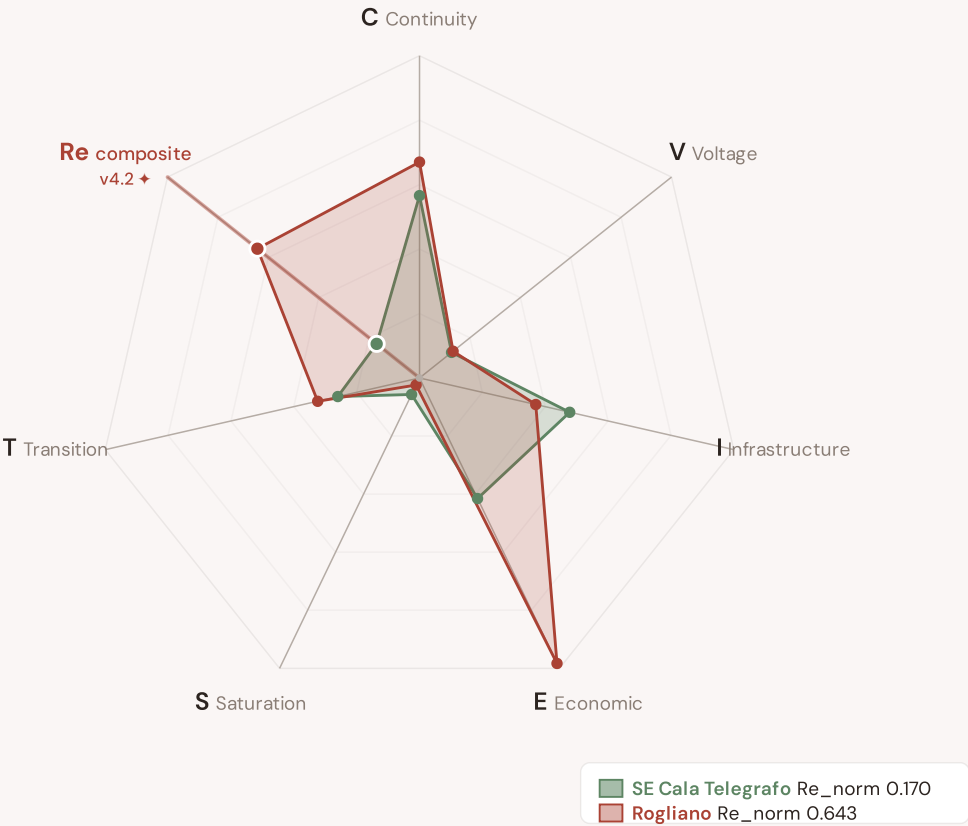
|                                            |                                                                         |
|--------------------------------------------|-------------------------------------------------------------------------|
| <b>A</b> Executive Summary                 | <b>E</b> European Context Card                                          |
| <b>B</b> Cyber & Economic Exposure Monitor | <b>F</b> SSI Enhanced Neural Network                                    |
| <b>C</b> Data Refresh & Changelog          | <b>F.5</b> Anti-maladaptation Gate (Radar 2 · W1–W10) <span>v4.2</span> |
| <b>D</b> Deep-Dive: The Puglia Corridor    | <b>G</b> Looking Ahead                                                  |

SECTION A

# Executive Summary

### Radar 1 — v4.2: now 7-axis with the new Re composite

v4.0.2 Radar 1 had 6 axes (C/V/I/E/S/T). **v4.2 adds Re — the bounded resilience composite** capturing the new modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just) in one normalised scalar ( $Re\_norm \in [0, 1]$  via  $clip((Re\_raw - 0.920) / (1.787 - 0.920), 0, 1)$ ). Two worked examples overlaid: **SE Cala Telegrafo** (low-exposure reference,  $Re\_norm$  0.170) · **Rogliano** (high-exposure reference,  $Re\_norm$  0.643). The Re axis is colour-highlighted in cayenne to emphasise the v4.2 architectural addition.



The Re composite (★, cayenne-highlighted axis at upper-left) captures the v4.2 resilience modifier family in one bounded scalar — formula  $Re\_raw = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$ , bounds  $[0.920, 1.787]$ . Per Convention #6 §2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are algorithmically selected (min/max  $Re\_norm$ ) per the v4.2 methodology. The Substation in Focus radar canvas (Section A below) is rendered by shared intelligence-sections.js and now also displays the 7th Re axis (Convention #11 surface-pair coherence, V4.2-disc-24 closure 12 Jun 2026).

#### SUBSTATIONS SCORED

4,293

475 EHV · 3,035 HV · 783 distribution-tier

#### GRID LINES MAPPED

14,221

~46,000 km coverage

#### MC SIMULATIONS

42.9 M

10,000 per substation

#### PUBLIC DATA SOURCES

34

101 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1

(48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 146,000+ source data points per run, executes 42.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 244,701 output data points with full uncertainty quantification across 4,293 substations and 14,221 grid lines spanning ~46,000 km. v4.2 adds the Re composite (7th axis) and six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just) on top of the v4.0.2 foundation.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 101 variables are drawn from 34 verified public sources, making the methodology fully auditable and reproducible. Initially covering Italy's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Substation 7000014126

ITH36, Veneto · 20 kV

0.511

Low

0.194

24th

R\_FINAL

CLASSIFICATION

CI WIDTH

FLEET PERCENTILE

Re composite v4.2 7th axis

Re\_raw 1.4491 · Re\_norm 0.610

ELEVATED EXPOSURE

COMPONENTS

C

Continuity: 0.353 / 0.30 (118%)

I

Infrastructure: 0.252 / 0.25 (101%)

S

Saturation: 0.444 / 0.20 (222%)

V

Voltage: 0.295 / 0.10 (295%)

E

Economic: 0.379 / 0.10 (379%)

T

Transition: 0.246 / 0.05 (493%)

MODIFIERS

R3

Consequence: 0.891 ↓ dampens

R4

Graph Criticality: 0.957 ↓ dampens

R6a

Restoration: 0.996 → neutral

R6b

Seismic: 1.050 ↑ amplifies

R7

Digital Readiness: 1.008 → neutral

This month's spotlight — automatically selected for August 2026 — is **Substation 7000014126** in ITH36, Veneto. With an R\_final of 0.511 (Low band, 24th fleet percentile), its risk profile is dominated by the T (Transition) component at 493% of its weight ceiling, followed by E (Economic) at 379%. Modifiers collectively shift R\_base by 51.5%%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

**Why this section exists:** Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that PNRR digitalisation planning requires.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

9982

High/Critical/Extreme + R7 < median

AVG R7 MODIFIER

1.0200

Range [0.99, 1.05]

HIGH-CONSEQUENCE SUBS

0

0.0% · R3 ≥ 1.16

### B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R\_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **9982 blind-spot substations** — scoring High, Critical, or Extreme on overall risk while sitting below the fleet median for digital readiness (R7 < 1.0201). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Puglia (2340)**, **Toscana (2103)**, **Friuli-Venezia Giulia (1535)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

### B.2 Economic Impact by Business Fabric

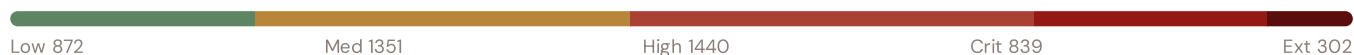
Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

### Capital-Intensive / High-Consequence

Est. VoLL: €15–30/kWh

Substations serving highest-value economic activity — pharma / chemicals / data centres / financial districts / heavy industry

4,804 substations (10.0%) High/Crit/Ext: **2581** (53.7%) Avg R: **0.613** Avg E: **0.444** / 0.10

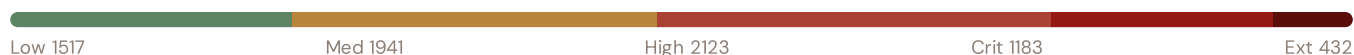


### Industrial / Medium-Large Enterprise

Est. VoLL: €8–15/kWh

Industrial belt — significant continuity requirements

7,196 substations (15.0%) High/Crit/Ext: **3738** (51.9%) Avg R: **0.608** Avg E: **0.418** / 0.10

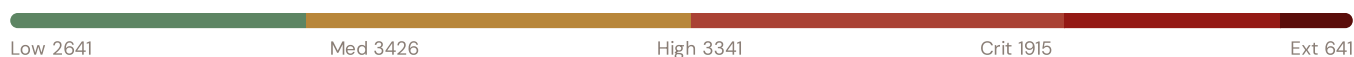


### Commercial / SME-Dense

Est. VoLL: €3–8/kWh

Service economy, retail, urban commercial districts — moderate individual impact

11,964 substations (25.0%) High/Crit/Ext: **5897** (49.3%) Avg R: **0.603** Avg E: **0.376** / 0.10

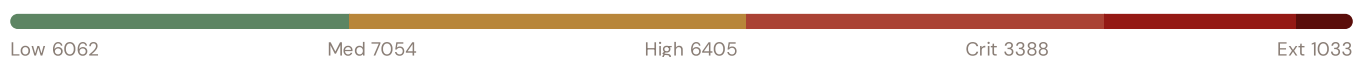


### Light / Agricultural / Rural

Est. VoLL: €1–3/kWh

Sparse activity, agricultural land — lower VoLL

23,942 substations (50.0%) High/Crit/Ext: **10826** (45.2%) Avg R: **0.593** Avg E: **0.297** / 0.10



**Value of Lost Load (VoLL) context:** ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's Economic component (CVIESTR-E) segments substations by the economic value they serve. **2581 substations** combine capital-intensive economic fabric (top-decile Economic component ( $E \geq 0.434$ , P90+)) with High, Critical, or Extreme risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Puglia (1207)**, **Toscana (923)**, **Friuli-Venezia Giulia (730)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 8.6%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

## B.3 Province Digital Readiness Ranking

Provinces ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R<sub>median</sub> to identify provinces combining digital exposure with high grid stress.



WORST DIGITAL READINESS — TOP 15 REGION REGIONS

Longer bar = higher R7 = better digital infrastructure.

|    |                             |             |        |
|----|-----------------------------|-------------|--------|
| 1  | ITF64 <span>▲ high R</span> | <div></div> | 1.0174 |
| 2  | ITF21                       | <div></div> | 1.0175 |
| 3  | ITF52                       | <div></div> | 1.0176 |
| 4  | ITG2H                       | <div></div> | 1.0183 |
| 5  | ITG2G                       | <div></div> | 1.0184 |
| 6  | ITF51 <span>▲ high R</span> | <div></div> | 1.0185 |
| 7  | ITG14                       | <div></div> | 1.0185 |
| 8  | ITC13 <span>▲ high R</span> | <div></div> | 1.0187 |
| 9  | ITG15                       | <div></div> | 1.0187 |
| 10 | ITI31 <span>▲ high R</span> | <div></div> | 1.0187 |
| 11 | ITC17                       | <div></div> | 1.0190 |
| 12 | ITH33 <span>▲ high R</span> | <div></div> | 1.0191 |
| 13 | ITC31 <span>▲ high R</span> | <div></div> | 1.0191 |
| 14 | ITG16 <span>▲ high R</span> | <div></div> | 1.0192 |
| 15 | ITC32 <span>▲ high R</span> | <div></div> | 1.0192 |

region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **ITF64** has the lowest average R7 (1.0174), while **ITI35** leads at 1.0281 — a spread of 0.0107 across the R7 range. **24 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

**Inaugural edition — Baseline Snapshot (August 2026).** This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0200, median = 1.0201; 0 substations classified HIGH cyber-exposure; 9982 blind-spot substations (High/Critical/Extreme risk + below-median R7); 3631 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **34 verified public data sources** feeding 101 variables across 6 components and 11 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

34

101 variables

HOURLY — MONTHLY

5

High-frequency feeds

QUARTERLY / ANNUAL

22

Regular refresh cycle

STATIC / DERIVED

# 8

Standards + scenarios

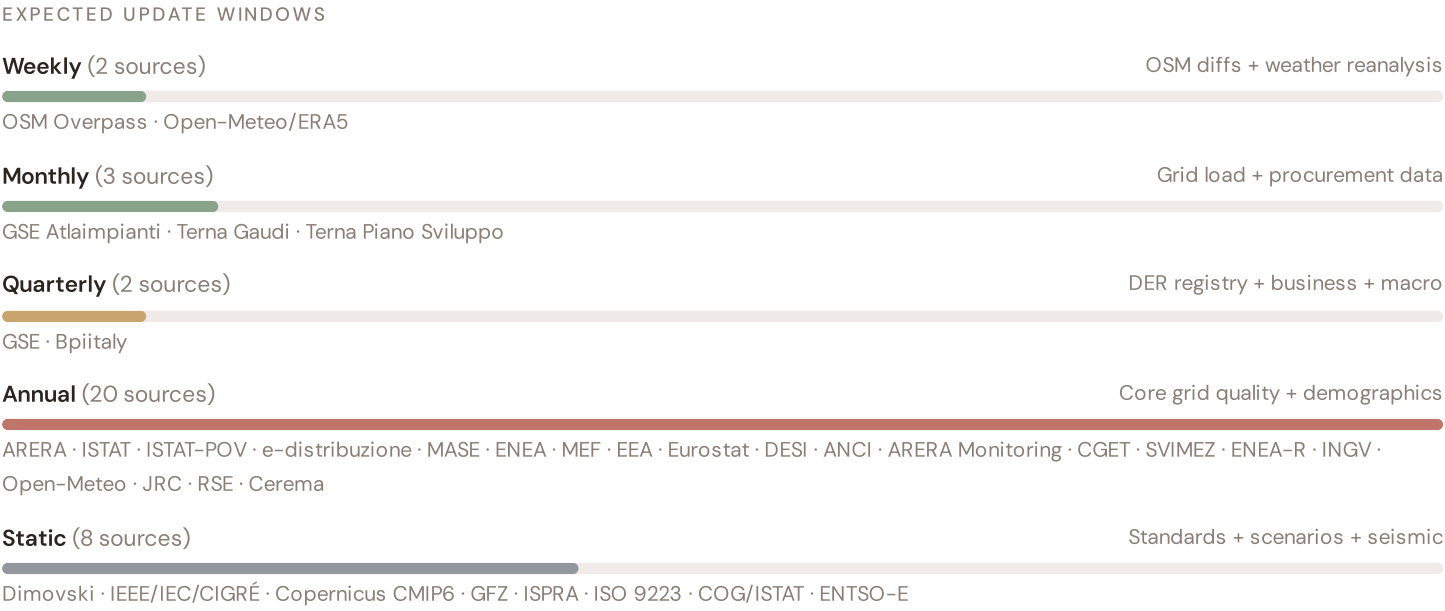
Data Source Registry — August 2026

|          |                                     |            |                         |        |
|----------|-------------------------------------|------------|-------------------------|--------|
| OSM      | OSM Power Infrastructure            | WEEKLY     | Node/edge               | 3 vars |
| AM       | Aeronautica Militare — Meteo        | DAILY      | ~1 km                   | 3 vars |
| TERNA    | Terna SpA (TSO)                     | MONTHLY    | Control area / zone     | 8 vars |
| INGV     | INGV — Geofisica e Vulcanologia     | CONTINUOUS | ~5 km                   | 4 vars |
| PCM-DPC  | Dipartimento Protezione Civile      | CONTINUOUS | Comune                  | 2 vars |
| COPER-EM | Copernicus Emergency Management     | CONTINUOUS | Event-based             | 1 var  |
| CSIRT    | CSIRT-Italia (ACN sub-org)          | CONTINUOUS | National                | 1 var  |
| GSE      | GSE Gestore Servizi Energetici      | QUARTERLY  | Regione                 | 4 vars |
| BdI      | Banca d'Italia                      | QUARTERLY  | Regione                 | 3 vars |
| ENEL-XR  | e-distribuzione (Enel DSO)          | QUARTERLY  | Provincia               | 3 vars |
| ARERA    | ARERA Regulator                     | ANNUAL     | DSO + Provincia         | 8 vars |
| ISTAT    | ISTAT — Italian National Statistics | ANNUAL     | Regione + Provincia ... | 9 vars |
| ISPRA    | ISPRA Environmental Protection      | ANNUAL     | Regione + bacino        | 4 vars |
| ACN      | ACN Cybersecurity Agency            | ANNUAL     | National + sector       | 2 vars |
| DESI     | DESI Digital Economy Index          | ANNUAL     | EU + NUTS-2             | 2 vars |
| EEA      | EEA Air Quality e-Reporting         | ANNUAL     | ~1 km + station         | 3 vars |
| EURO     | Eurostat Energy Statistics          | ANNUAL     | NUTS-2                  | 3 vars |
| SNAM     | SNAM Gas Network Operator           | ANNUAL     | National                | 1 var  |
| RSE      | RSE Ricerca sul Sistema Energetico  | ANNUAL     | Regione                 | 2 vars |
| MIT      | MIT — Ministero Infrastrutture      | ANNUAL     | National                | 1 var  |
| AUTBAC   | Autorità di Bacino Distrettuali     | ANNUAL     | 7 bacini                | 2 vars |
| JRC      | JRC DSO Observatory                 | ANNUAL     | DSO-level               | 2 vars |
| ARETI    | Areti (ACEA Group, Roma DSO)        | ANNUAL     | Comuni Roma             | 2 vars |
| UNARETI  | Unareti (A2A Group, Milano DSO)     | ANNUAL     | Milano + Brescia        | 2 vars |



|           |                                    |         |                   |         |
|-----------|------------------------------------|---------|-------------------|---------|
| INRETE    | Inrete (Hera Group, ER DSO)        | ANNUAL  | Emilia-Romagna    | 2 vars  |
| SETDIS    | Set Distribuzione (Trentino)       | ANNUAL  | Trentino-AA       | 1 var   |
| ISPRA-INV | ISPRA IFFI Landslide Inventory     | ANNUAL  | Comune            | 1 var   |
| ANAC      | ANAC Anti-Corruption Authority     | ANNUAL  | National          | 1 var   |
| D10       | Copernicus CDS / ERA5              | STATIC  | 0.25° (~25 km)    | 4 vars  |
| IEEE-1    | IEEE C57.91 / IEC 60076            | STATIC  | Asset-level       | 16 vars |
| CIGRE     | CIGRE TB 761 Asset Management      | STATIC  | Asset-level       | 5 vars  |
| GFZ       | GFZ German Research Centre         | STATIC  | ~5 km             | 2 vars  |
| ISO-9223  | ISO 9223 Corrosion                 | DERIVED | Regione + coastal | 1 var   |
| GME       | GME Gestore dei Mercati Energetici | HOURLY  | 7 bidding zones   | 3 vars  |
| ENTSE     | ENTSO-E Transparency               | HOURLY  | Bidding zone      | 2 vars  |

C.1 Update Frequency Timeline



C.2 Fleet Score Snapshot

FLEET SIZE

47,906

substations scored

MEDIAN R

0.598

P5=0.454 · P95=0.753

HIGH + CRITICAL + EXTREME

23042

48.1% of fleet

HIGH CONFIDENCE

96%

46169 subs · CI/R &lt; 0.50

## BAND DISTRIBUTION



No primary data sources have been updated since the v4.0.2 launch baseline. All **47,906 substation scores remain unchanged** this edition. The fleet median R of 0.598 (mean 0.600) reflects a distribution skewed toward the lower bands, with 23.2% in Low and 28.7% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

### C.3 Changelog — v3.4 → v4.0.2 → v4.2

v4.2 (June 2026) adds six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just), the Re composite (7th axis, bounds [0.920, 1.787]), four new data sources (ISPRA IdroGEO PGRA, ISPRA SIAB wildfire, ECMWF ERA5 Bourgouin, ARERA bulk smart meter), six new variables, and W1–W10 anti-maladaptation governance gate (5/1/1/3 4-tier mesh resolution per Convention #6 §5.6) on top of the v4.0.2 baseline.

27 changes tracked across PO, P1, and P2 releases

|        |          |                                                                                                                                                                               |          |
|--------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| V42-1  | NEW      | R6c_flood — PGRA flood vulnerability modifier (ISPRA IdroGEO, additive cliff +0.00..+0.30, P3=high/P2=medium/P1=low per Italian PGRA convention)                              | \$5      |
| V42-2  | NEW      | R6d_wildfire — Canadian Fire Weather Index modifier (EFFIS FWI + NASA FIRMS + ISPRA SIAB national-tier, multiplicative [1.00, 1.20])                                          | \$5      |
| V42-3  | NEW      | R6e_winter — Bourgouin freezing-rain p99 tail-event amplifier (ERA5, multiplicative [1.00, 1.15], Italy collapses to ~1.005 per brief §6 methodologically-defensible absence) | \$5      |
| V42-4  | NEW      | R8_adapt — Adaptive capacity (ARERA bulk SM + ENEA + DESI, multiplicative)                                                                                                    | \$5      |
| V42-5  | NEW      | R9_compound — Pairwise hazard coupling (R6b×R6c, R6c×R6d, R6c×R6e, multiplicative)                                                                                            | \$5      |
| V42-6  | NEW      | R10_just — Energy justice (EU-SILC + OIPE energy-poverty, multiplicative)                                                                                                     | \$5      |
| V42-7  | NEW      | Re composite — Resilience composite (R6d × R6e × R8 × R9 × R10) + (R6c – 1.00); bounds [0.920, 1.787]                                                                         | \$5b     |
| V42-8  | DATA     | 101 variables (was 95) — +6 modifier inputs from new resilience family                                                                                                        | \$8      |
| V42-9  | DATA     | 34 data sources (was 30) — +4: ISPRA IdroGEO PGRA, ISPRA SIAB+Carabinieri, ECMWF ERA5 Bourgouin, ARERA bulk SM                                                                | \$12     |
| V42-10 | NEW      | W1–W10 5/1/1/3 4-tier audit-state split (5 Published / 1 Published–baseline ▲ / 1 Baseline-only / 3 Commercial) — mesh resolution per Convention #6 §5.6                      | W1–W10   |
| F1     | NEW      | New T component — Energy Transition Exposure (T1)                                                                                                                             | \$2, \$4 |
| F2     | NEW      | New R6a — Restoration Speed Modifier (CAIDI-based)                                                                                                                            | \$5      |
| F3     | ENHANCED | R3 enhanced — Energy Poverty Vulnerability (V_socio)                                                                                                                          | \$5      |
| F4     | ENHANCED | R4 enhanced — Graph-theoretic betweenness + bridge detection                                                                                                                  | \$5      |

|    |          |                                                                                                             |           |
|----|----------|-------------------------------------------------------------------------------------------------------------|-----------|
| F5 | ENHANCED | R2 enhanced — Climate Trajectory (CMIP6 SSP2–4.5)                                                           | \$5       |
| F6 | ENHANCED | R7 Digital Readiness — Provincia-level DESI model                                                           | \$5       |
| L1 | NEW      | New I5, I7–I9 metrics — thermal, load, corrosion, hydrogeo                                                  | \$2, \$8  |
| L2 | ENHANCED | E2 Innovation Enrichment — HRST + startup density                                                           | \$6       |
| L3 | ENHANCED | V_socio Fiscal Enrichment — MEF + ANCI + energy price                                                       | \$5       |
| L4 | ENHANCED | R3 Demographic Shift Amplifier — ISTAT net migration                                                        | \$5       |
| L5 | DATA     | 95/95 variables operational (v4.0.2 baseline; 101/101 in v4.2). 30 data sources total (v4.0.2; 34 in v4.2). | \$8, \$12 |
| G1 | DATA     | d02 GSE Atlaimpianti upgraded to quarterly registry — Provincia-level DER registry                          | \$12      |
| G2 | DATA     | d05 ISPRA upgraded to live API — Provincia-level geological data                                            | \$12      |
| G3 | DATA     | d06 OSM upgraded to Overpass API — 7,898 real substations                                                   | \$12      |
| G4 | NEW      | R6b Network Topology modifier — centrality + ring analysis from OSM graph                                   | \$5       |
| G5 | NEW      | Network & Topology layer (H): 7 variables — Terna Piano di Sviluppo, ring analysis                          | \$12      |
| G6 | NEW      | Output Scores layer (I): 7 variables — risk_score, ETTC, stationary probs                                   | \$12      |

SECTION D — MONTHLY DEEP-DIVE

# The Puglia Corridor: Where the Energy Transition Is Outrunning the Grid

**Deep-dive rotation:** Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Puglia corridor · **002** DER saturation hotspots · **003** Climate vulnerability under CMIP6 · **004** North–South economic impact disparity · **005** Voltage quality and industrial output · **006** Infrastructure age vs investment · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** Provincial Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

## D.1 Why Puglia, Why Now

PUGLIA SUBSTATIONS

435

202 EHV · 233 HV

HIGH BAND

127 + 80 + 57

60.7% of corridor

CORRIDOR MEDIAN R

0.634

Fleet median: 0.598

WORST PROVINCE

ITF64

Avg R: 0.678 · 46 substations

The Calabria hosts **435 substations** across 5 region regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **127 substations (29.2%)** are classified High and **80 Critical**, with only **69** in the Low band. 62% of corridor substations score above the national fleet median of 0.598. The corridor median R of 0.634 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring



| SUBSTATION               | PROVINCE | R_FINAL | BAND    | C     | V     | I     | E     | S     | T     | RE<br>v4.2 | ALERT |
|--------------------------|----------|---------|---------|-------|-------|-------|-------|-------|-------|------------|-------|
| Substation<br>7000049392 | ITF63    | 1.012   | Extreme | 0.430 | 0.349 | 0.334 | 0.311 | 0.432 | 0.293 |            |       |
| Maida                    | ITF63    | 0.927   | Extreme | 0.318 | 0.350 | 0.306 | 0.360 | 0.441 | 0.415 |            |       |
| Andali                   | ITF63    | 0.906   | Extreme | 0.432 | 0.438 | 0.297 | 0.309 | 0.261 | 0.303 |            |       |
| Substation<br>7000049535 | ITF61    | 0.901   | Extreme | 0.427 | 0.371 | 0.447 | 0.371 | 0.295 | 0.278 |            |       |
| Substation<br>7000049603 | ITF64    | 0.891   | Extreme | 0.444 | 0.418 | 0.447 | 0.276 | 0.449 | 0.357 |            |       |
| Substation<br>7000049543 | ITF64    | 0.870   | Extreme | 0.326 | 0.453 | 0.434 | 0.268 | 0.420 | 0.266 |            |       |
| Substation<br>7000049345 | ITF61    | 0.850   | Extreme | 0.413 | 0.435 | 0.254 | 0.430 | 0.431 | 0.372 |            |       |
| Substation<br>7000049571 | ITF65    | 0.850   | Extreme | 0.414 | 0.343 | 0.365 | 0.380 | 0.372 | 0.409 |            |       |

| SUBSTATION               | PROVINCE | R_FINAL | BAND    | C     | V     | I     | E     | S     | T     | RE<br>v4.2 | ALERT |
|--------------------------|----------|---------|---------|-------|-------|-------|-------|-------|-------|------------|-------|
| Substation<br>7000049568 | ITF65    | 0.849   | Extreme | 0.277 | 0.446 | 0.395 | 0.324 | 0.278 | 0.380 |            |       |
| Substation<br>7000049618 | ITF64    | 0.849   | Extreme | 0.377 | 0.426 | 0.430 | 0.363 | 0.429 | 0.367 |            |       |

... 425 additional substations — [explore full corridor in Digital Twin →](#)

### D.3 Component Analysis

COMPONENT AVERAGES — CALABRIA VS FLEET

|                    |                      |
|--------------------|----------------------|
| C — Continuity     | 0.346 vs 0.350 (−1%) |
| I — Infrastructure | 0.358 vs 0.350 (+2%) |
| S — Saturation     | 0.347 vs 0.350 (−1%) |
| V — Voltage        | 0.352 vs 0.350 (0%)  |
| E — Economic       | 0.343 vs 0.350 (−2%) |
| T — Transition     | 0.347 vs 0.349 (−1%) |

MODIFIER AVERAGES — CALABRIA VS FLEET

|                  |       |               |
|------------------|-------|---------------|
| R3 Consequence   | 0.891 | fleet 0.891 ↓ |
| R4 Graph Crit.   | 0.980 | fleet 0.980 ↓ |
| R6a Restoration  | 1.020 | fleet 1.020 ↑ |
| R6b Seismic      | 1.138 | fleet 1.052 ↑ |
| R7 Digital Read. | 1.020 | fleet 1.020 ↑ |

The dominant risk driver across the Calabria corridor is **T (Transition)**, averaging 0.347 against a fleet average of 0.349 — occupying 694% of its weight ceiling (w = 0.05). The second contributor is **V (Voltage)** at 0.352 vs fleet 0.350 (352% of ceiling). Among modifiers, the R3 consequence multiplier averages 0.891 (fleet: 0.891), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.138, capturing the local seismic exposure — a dimension invisible to every competing framework.

### D.4 Province Breakdown

REGION RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

|       |                                                |
|-------|------------------------------------------------|
| ITF64 | 46 subs · avg R = 0.678 · H:20 C:5 E:12 M:8    |
| ITF63 | 62 subs · avg R = 0.659 · H:19 C:10 E:13 M:11  |
| ITF65 | 92 subs · avg R = 0.658 · H:28 C:26 E:12 M:20  |
| ITF61 | 183 subs · avg R = 0.612 · H:50 C:29 E:16 M:50 |
| ITF62 | 52 subs · avg R = 0.599 · H:10 C:10 E:4 M:13   |

The region regions-level breakdown reveals significant intra-regional variation. **ITF64** leads with a mean R of 0.678 across 46 substations, while **ITF62** scores 0.599 — a spread of 0.079 within a single corridor. The SSI's substation-level granularity reveals that the Calabria is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

**191 Calabria substations** score  $R \geq 0.650$ , placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the ITF64 sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

SECTION E — RECURRING MODULE

European Context Card

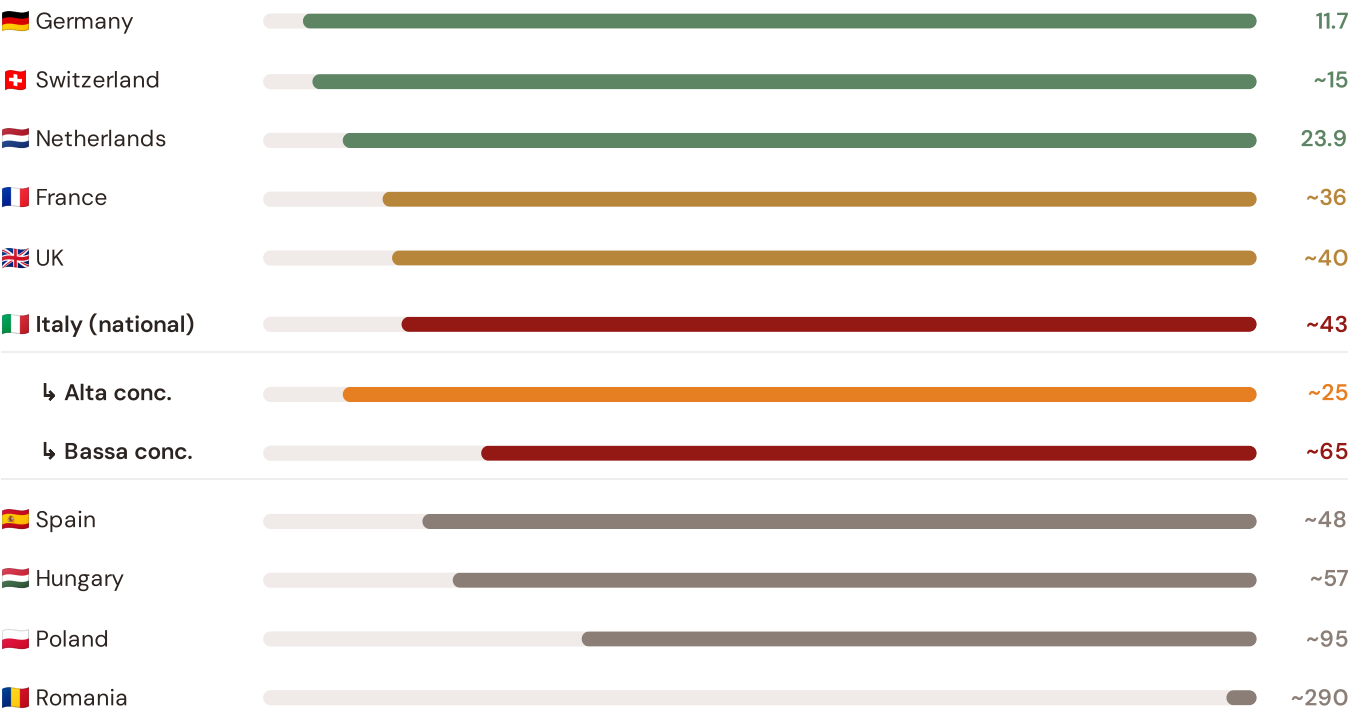
**How this works:** Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Puglia corridor analysis hinges on Italy's continuity performance relative to European peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

EUROPEAN CONTEXT CARD — SAIDI BENCHMARKING

Ed. 001 · August 2026

Unplanned SAIDI (min/customer/year) — Selected Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



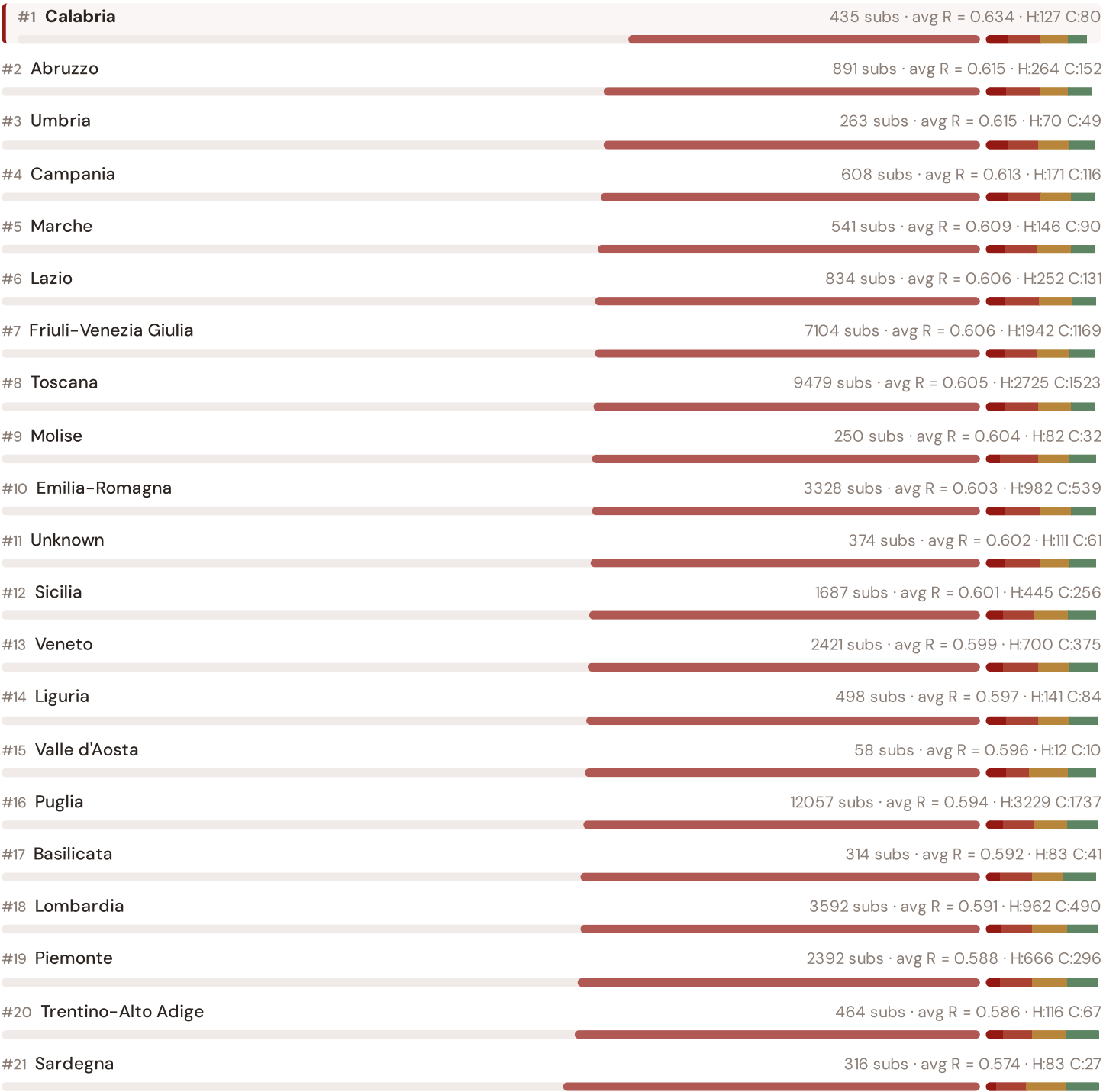
Source: CEER 7th Benchmarking Report (2022); Bundesnetzagentur (2024); ARERA (2023). Italy sub-categories from ARERA territory classification. · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release.

**This month's key insight:** The Calabria corridor deep-dive shows **435 substations** (of 435) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.346 is **1.0× the fleet average** of 0.350. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

## Regional Risk Ranking — All 20 Italian Regions

Where does Puglia sit within the national landscape? Regions sorted by mean R<sub>median</sub>, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



The Calabria ranks **#1 of 21 region regions** by mean R<sub>median</sub>, placing it among the upper tier of national risk. The three most stressed are Calabria, Abruzzo, Umbria, while the three most resilient are Sardegna, Trentino-Alto Adige, Piemonte. 12 of 21 score above the fleet average of 0.600. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this

substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

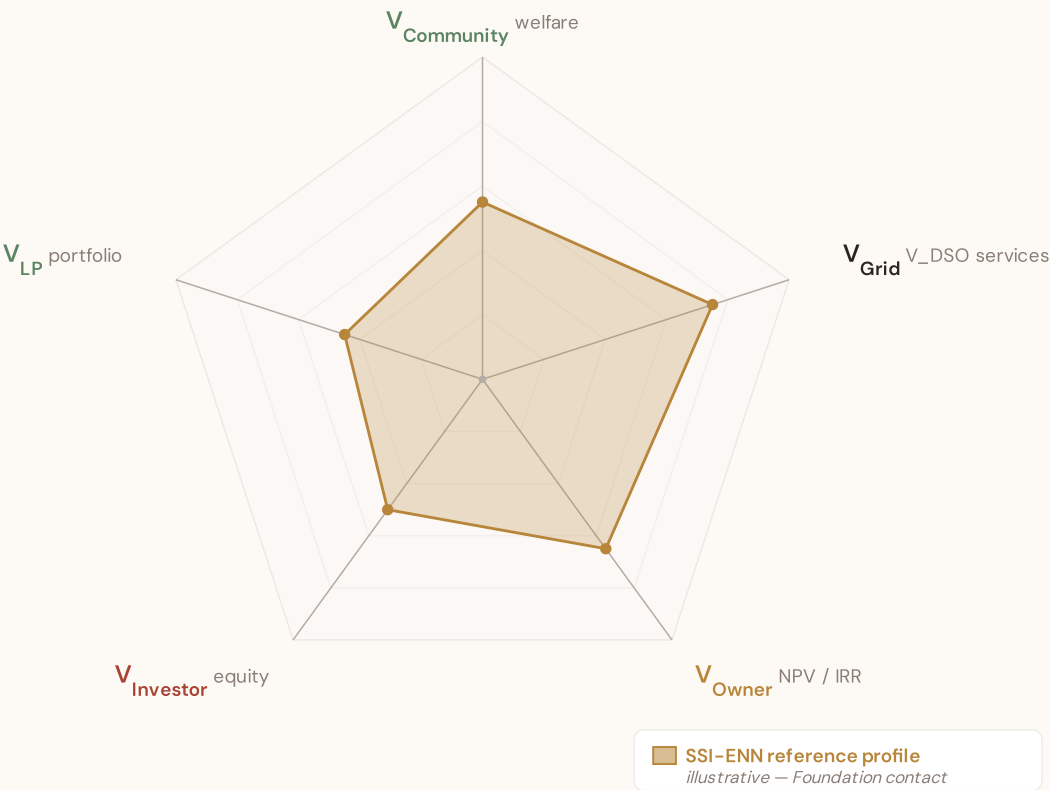
SECTION F — RECURRING MODULE

# SSI Enhanced Neural Network — Monthly Topic

**From grid intelligence to BESS valuation platform.** The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the DNO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

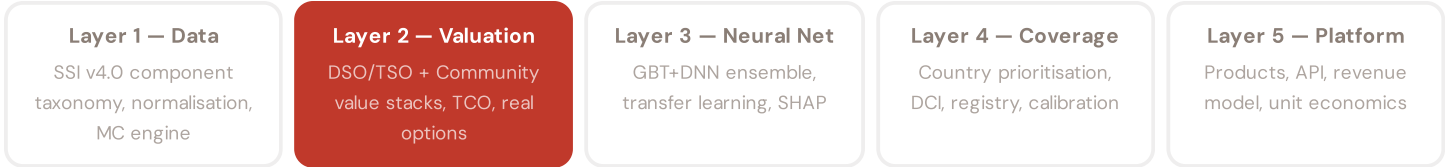
## SSI-ENN value-articulation radar — 5-lens stakeholder taxonomy

The SSI-ENN's commercial-tier value articulation answers "what is a BESS worth at this substation — and to whom?" through five stakeholder lenses (architecture memo §5 Decision #12): **V<sub>Community</sub>** (welfare avoided outages, energy-poverty relief, employment) · **V<sub>Grid</sub>** (V\_DSO — avoided reinforcement, reliability, voltage, congestion, frequency, black-start, hosting capacity) · **V<sub>Owner</sub>** (NPV / IRR / payback per asset class) · **V<sub>Investor</sub>** (risk-adjusted equity return at the financing tier) · **V<sub>LP</sub>** (limited-partner cash-on-cash + DPI / TVPI portfolio metrics). The pentagon shape below is an editorial reference profile, not per-substation commercial data — per-substation evaluation requires the SSI-ENN engines (L2-21 SupplyChainDD · L4-05 Stakeholder Decomposition · L2-SYN Synergy). Foundation contact: [ssi\\_index@ikenga.eu](mailto:ssi_index@ikenga.eu).



*Editorial reference shape — V<sub>Grid</sub> dominates (V\_DSO captures avoided reinforcement + reliability + voltage support + congestion relief + frequency + black-start + hosting capacity, the bulk of measurable BESS value at the substation locus per architecture memo §5). V<sub>Community</sub> + V<sub>Owner</sub> are co-secondary. V<sub>Investor</sub> + V<sub>LP</sub> reflect the financing-tier waterfall above V<sub>Owner</sub> NPV. Per-substation commercial decomposition lives in the SSI-ENN — mail to [ssi\\_index@ikenga.eu](mailto:ssi_index@ikenga.eu) for analytics.*





LAYER 2 §0.4.3 · August 2026

## Total Cost of Ownership Model

*Stochastic cost modelling — because BESS economics are not deterministic*

CAPEX follows a learning-curve distribution calibrated to Bloomberg NEF projections. OPEX is modelled as a mean-reverting process with seasonal components reflecting temperature-dependent degradation. Battery degradation follows a non-linear capacity fade model: each cycle reduces usable capacity, with rate depending on depth-of-discharge, temperature, and C-rate. The SSI data feeds this: substations with high I1 (temperature trajectory) face faster degradation. The Monte Carlo engine propagates cost uncertainty alongside revenue uncertainty.

### KEY CONCEPTS

- CAPEX learning curve calibrated to BNEF
- Non-linear battery degradation:  $f(\text{DoD}, \text{temperature}, \text{C-rate})$
- SSI I1 (climate trajectory) drives degradation forecast
- Full NPV distribution, not point estimate

### LIVE SSI DATA CONNECTION

Across the 47,906-substation fleet, the average consequence multiplier (R3) is 0.891, indicating significant socio-economic exposure. 23042 substations (48.1%) are classified High or Critical — these are the highest-priority candidates for BESS deployment.

**Coming next month:** **LAYER 1** Gaussian Copula Monte Carlo Engine — *How 10,000 simulations per substation capture the correlation structure across grid risk factors*

## SECTION F.5 — V4.2 ANTI-MALADAPTATION GOVERNANCE GATE

## Radar 2 — W1–W10 Substation Community Value

**Wall 1 — Public W1–W10 framework.** The 10-axis Radar 2 expresses per-substation community-value baseline pre-intervention: *W1 Reliability · W2 Carbon · W3 Local Economic · W4 Climate Resilience · W5 Digital Inclusion · W6 Sovereignty · W7 Local Value Retention · W8 Communications · W9 Supply Security · W10 Compute Multiplier*. The discipline floor that the anti-maladaptation governance gate protects: *"no W axis worse off after intervention."*

5/1/1/3 4-tier audit-state classification (mesh resolution per Convention #6 §5.6)

Published (5) — solid fill

W1 · W2 · W3 · W4 · W8

Peer-reviewed methodology operational at fleet scale.

Published-baseline (1) ▲ — solid fill + chevron corner marker

W9 — Supply Security

Peer-reviewed academic basis (Albert/Holme topology-only baseline) with acknowledged scope limit (Buldyrev/IEEE 493 — N-1 contingency dynamics not captured at baseline).

Baseline-only (1) — striped fill

W5 — Digital Inclusion

Foundation methodology under development; baseline value defensible at v4.2.

Commercial (3) — dashed-faded

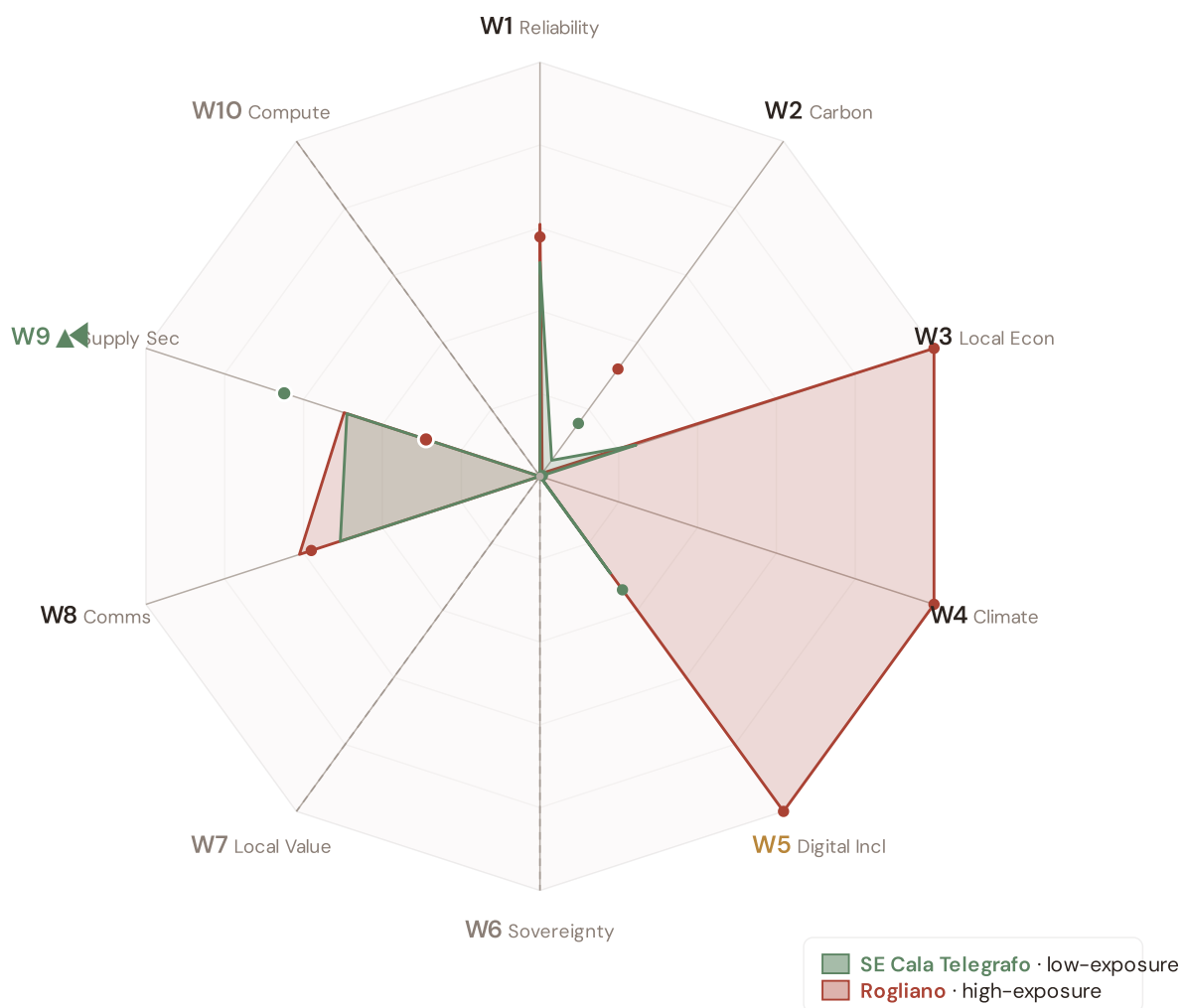
W6 · W7 · W10

Commercial deliverables — please contact the Foundation.

Per Convention #6 §5.6.7 academic-defensibility: the 4-tier classification aligns with IPCC AR6 + JRC COIN + Reckien (2021) continuum literature. The Published-baseline tier (W9) is the principled mesh resolution that honors both peer-review status + acknowledged scope limit without forcing a false binary.

## Radar 2 visualization — anti-maladaptation governance gate

Two worked examples overlaid: **SE Cala Telegrafo** (low-exposure reference,  $Re\_norm$  0.170) and **Rogliano** (high-exposure reference,  $Re\_norm$  0.643). Axis labels are colour-coded per the 5/1/1/3 mesh resolution; the ▲ chevron marks W9 as Published-baseline. Hover any axis label for the methodology tooltip.



*SE Cala Telegrafo polygon (sage): the low-exposure reference — compact footprint reflects this country's fleet floor on the W-axis baselines.*  
*Rogliano polygon (terracotta): the high-exposure reference — larger footprint where this country's fleet's W-axis values reach the upper percentile. Commercial-tier axes (W6 data sovereignty, W7 local value retention, W10 compute multiplier) display at zero per Convention #6 §5.6 mesh design — no public-data computation. W2 carbon uses Italy-specific zonal MEF per ssi\_enn\_v30 v4.2 calibration.*

W1–W10 axis enumeration with audit-state classification

5/1/1/3 4-TIER · V4.2 MESH RESOLUTION · LIVE VALUES VIA COMPUTE\_W\_BASELINES.PY

| AXIS | NAME                  | AUDIT STATE          | METHODOLOGY BASIS                                                                                                                                                                                             |
|------|-----------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| W1   | Reliability           | Published            | $C \times E2_{\text{local}} \times \text{population}$ (ARERA SAIDI + ISTAT + ISTAT-POV)                                                                                                                       |
| W2   | Carbon                | Published            | $T1 \times \text{zonal MEF} \times (V \cdot \text{deg} / 100)$ (Terna + EEA)                                                                                                                                  |
| W3   | Local Economic        | Published            | $\text{norm}(R10_{\text{just}}) - \text{EU-SILC} + \text{OIPE energy-poverty}$                                                                                                                                |
| W4   | Climate Resilience    | Published            | $(R6d - 1) + (R6e - 1) + (R6c - 1) + \frac{1}{2}(R9 - 1) - \text{v4.2 resilience modifier family}$                                                                                                            |
| W5   | Digital Inclusion     | Baseline-only        | $\text{norm}(R8_{\text{adapt}}) - \text{ARERA bulk SM} + \text{ENEA} + \text{DESI}$                                                                                                                           |
| W6   | Sovereignty           | Commercial           | Foundation contact ( <a href="mailto:ssi_index@ikenga.eu">ssi_index@ikenga.eu</a> )                                                                                                                           |
| W7   | Local Value Retention | Commercial           | Foundation contact ( <a href="mailto:ssi_index@ikenga.eu">ssi_index@ikenga.eu</a> )                                                                                                                           |
| W8   | Communications        | Published            | $C \times p_{\text{crit\_10yr}} \times \text{population}$ (Markov 10-yr critical probability)                                                                                                                 |
| W9 ▲ | Supply Security       | Published-baseline ▲ | $0.60 \times BC_{\text{pct}} + 0.30 \times \text{bridge} + 0.10 \times (1 - \text{deg} / 20) - \text{OSM power-graph betweenness centrality} + \text{bridge identification} + \text{degree-based redundancy}$ |
| W10  | Compute Multiplier    | Commercial           | Foundation contact ( <a href="mailto:ssi_index@ikenga.eu">ssi_index@ikenga.eu</a> )                                                                                                                           |

▲ W9 Published-baseline — mandatory transparency tooltip

W9 — Supply Security (Published-baseline). Public-tier methodology: betweenness centrality + bridge identification + degree-based redundancy, computed deterministically from OpenStreetMap power-graph data. Peer-reviewed academic basis: Albert et al. (2000) *Nature*; Holme et al. (2002); broad power-grid centrality literature. Acknowledged scope limit (per Buldyrev et al. (2010) *Nature* + IEEE 493): topology-only baseline does not capture N-1 contingency dynamics or interdependent-network cascades. Richer methodology — N-1 contingency simulation + flexibility-services valuation — lives in SSI-ENN L2-22 / L2-23 commercial deliverables. Mail to [ssi\\_index@ikenga.eu](mailto:ssi_index@ikenga.eu) for extended analysis.

**Worked examples.** The W1–W10 per-substation baseline values for two anchor cases — SE Cala Telegrafo (low-exposure reference) and Rogliano (high-exposure reference) — are computed live by `_audit-snapshot/scripts/d3_radar2_per_country.py` against the country's `ssi-data.json`. Per Convention #6 §2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are now algorithmically selected (min/max `Re_norm`) rather than named picks. See the FC v2 reference for W-axis methodology + commercial-tier rationale.

**Anti-maladaptation discipline.** The 10-axis Radar 2 baseline is the per-substation floor that any post-intervention scenario must protect. No W axis may degrade after a BESS deployment, line upgrade, or operational change. The discipline is enforced per axis (not in aggregate), and per substation (not in fleet average). This is the v4.2 governance gate that ensures grid-modernisation projects deliver net community value without hidden trade-offs.

SECTION G

Looking Ahead

NEXT MONTH — EDITION 04

Alpine Valleys

Deep-dive into Region 05's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Narrow valleys, long restoration MTTR.

NEXT DATA REFRESH

Q4 2026 — 2 Quarterly Sources

**GSE, Bpiitaly** are due for refresh in Q4. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **ISTAT — Italian National Statistics** (9 variables → R3 population, elderly, GDP, Mezzogiorno North-South split, fiscal capacity).

FLEET SPOTLIGHT

o Bridge Substations — Single Points of Failure

**O substations** are topological bridges — their failure disconnects part of the network. Of these, **O** are classified High or Critical. These are the fleet's single points of failure: high graph centrality (R4) combined with elevated risk makes them priority candidates for both grid reinforcement and BESS deployment.

Edition 04 will be published on the second Thursday of September 2026. Deep-dive region: **Region 05**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to [ssi\\_index@ikenga.eu](mailto:ssi_index@ikenga.eu).

**Feedback welcome.** The SSI Index is designed to evolve through stakeholder dialogue. If you represent an Italian utility, ARERA, municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at [ssi\\_index@ikenga.eu](mailto:ssi_index@ikenga.eu). The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 006

Published 13 August 2026 · SSI Index **v4.2** (refreshed June 2026) · 4,293 substations · 14,221 grid lines · 6 components + **Re composite** (Radar 17th axis, bounds [0.920, 1.787]) · 20 metrics · 11 modifiers (5 baseline + 6 v4.2 resilience family)

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5 · W1-W10 anti-maladaptation governance gate (5/1/1/3 4-tier mesh resolution per Convention #6 §5.6)

### Foundation contact

The SSI Index is a non-profit foundation project, please mail to [ssi\\_index@ikenga.eu](mailto:ssi_index@ikenga.eu) for more details.

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